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The case for neuropsychanalysis

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THE BRAIN is wider than the sky,
For, put them side by side,
The one the other will include
With ease, and you beside.

The brain is deeper than the sea,
For, hold them, blue to blue,
The one the other will absorb,
As sponges, buckets do.

The brain is just the weight of God,
For, lift them, pound for pound,
And they will differ, if they do,
As syllable from sound.

Emily Dickinson(1830–86)
Complete Poems (1924).
Part One: Life CXXXVI

Among a number of aims pursued by the authors of this paper, one was to analyse and discuss the objections and criticisms that have been made towards the discipline founded by the same authors – namely neuropsychanalysis.

My commentary will not focus on these criticisms, which are well known for that matter, but I will propose a set of observations somewhat ‘tangential’ to the critical debate and to the paper itself. I will strive to do this specifically in relation to the issues that I deem central in what, in the 2014 Symposium organised by the European Psychoanalytical Federation in Berlin (Canestri, 2014), I referred to as “The complexity of the dialogue between neurosciences and psychoanalysis”. It goes without saying that in this brief summary I will not be able to do justice to the authors’ well-structured and detailed works, and that I will have no choice but to repeat some of the arguments already developed in the Symposium (Canestri, 2014).

The importance of the premises or assumptions

The paper considers the early history of the relations between neuroscience¹ and psychoanalysis, highlighting the importance of considering the scientific

¹I would like to clarify that I will use this term in the wider possible sense, taking into account that this designation applies to a vast array of sub-disciplines, not all necessarily related to ‘neuropsychanalysis’.

premises of Freud's early writings on neurology. Otherwise, it would not be possible to properly comment on Solms and Saling's (1986, 1990) work on Freud's early writings.

In his formation of an independent discipline, Freud never abandoned his idea that his models of the mind could one day to *some extent* correlate with other models of brain functioning that had not yet been developed. The same could be said with regards to the open relations Freud always sought with biology.

The authors, in my view very appropriately, preface their paper by stating that certain arguments can only be set on constructive paths if the participants in the debates are ready to declare their starting assumptions. One such assumption that needs examining is the question – is psychoanalysis a science? Then there is the question of the definition of science and of which science is adopted. One must also consider Freud's positivism and mechanisms, the relationship between the mind and the brain, and meaning and the body... The list could be rather long.

Neuropsychanalysis was born in the attempt – made explicit and argued clearly in the article – to put the disciplines of neuroscience and psychoanalysis under scrutiny and in dialogue, as well as place the footprint of the psychoanalytical model onto neuroscientific research methodologies.

The article also recounts how we went from the original use of the dash between neuro and psychoanalysis to the current integration of both words in one. The matter may seem irrelevant, but perhaps it is not. The integration appears in this way to lead to the creation of a new discipline, an interdisciplinary approach.

But does actual interdisciplinarity apply in this context? Here we also face the need to define the assumptions: that is, define interdisciplinarity.

My premise is that one can talk of interdisciplinarity when two disciplines share the same methodology, language and have one epistemology. There are many disciplines that are integrated into one name in the configuration of a new discipline: astrophysics, biochemistry, biophysics, etc.

Neuroscience and psychoanalysis neither share a methodology nor a language, and the crossing over of data from each is not linear – that is, you cannot compare data as if it were identical in both disciplines. Perhaps, from this perspective, it would be better for one not to be reduced to the other, at a linguistic level too (with neuro-psychoanalysis turned to neuropsychanalysis). But this does not mean that there should be no dialogue between the two disciplines – the potential richness and reciprocal utility is not put into question.

By virtue of these considerations I prefer to talk about 'intersections' between disciplines which are diverse in terms of language, methodology and epistemology. The intersections may be more or less important, but we shouldn't apriori have the authority to reject one discipline or declare it irrelevant.

The issue of relevance and the associated epistemological problem

If we adopt a post-empirical epistemology, refuting Popper's (1934) classical distinctions between context/ or the logic of discovery and context/ or the logic of demonstration, and if we embrace Lakatos's (1976) suggestion that you can't make a clear distinction between the history of a discipline and the discipline itself, then we can argue in favour of a heuristic course of the theory.

It can be demonstrated, in our case, that the relationship between the history and the discipline is clearly essential to Freudian theory. We have to add that what Popper called *background knowledge* has to be interpreted within this framework as the whole set of facts used in the construction of a specific theory, beyond the understanding available at the time of its formulation (Mottierlini, 1964; Hesse, 1966; Canestri, 2006, 2014). A three-way relationship between evidence, theory and heuristic progress can be conceived as a *theory-arrived-at-in-a-certain-way*, namely a *theory with its heuristics*.

Some well-known examples could show how in his constructions Freud, like many thinkers, used different types of theoretical materials stemming from multiple disciplines (neurology, physics, philosophy of language, etc.). These materials transform themselves when they are included in and adapted to new theoretical configurations; this takes place, for instance, in Freud's theory of affect, in his concepts of representations and drive, and his theory of aphasia, etc.

I have to add another consideration. In her analysis of the theory of models, Mary Hesse (1966) underlines their essential role in scientific practice and the heuristic value of models. A *theoretical model* in the strict sense refers to "a collection of assumptions about an object or system that describe it by attributing to it an 'internal' structure so that many of its properties are explained by referring to this structure" (Hesse, 1966:8). An *imaginary model* is "a collection of assumptions about an object or system that show us what it could be if it satisfied certain conditions which in fact it does not satisfy" (Hesse, 1966:8).

It is reasonable to think that in the instances outlined above Freud may have made use of imaginary models of great heuristic power, which however did not strictly satisfy the conditions imposed by the theoretical models. The concept of psychic energy, for instance, surely did not satisfy the conditions set forth by Helmholtz's (1967) model. Freud was fully aware of this fact whenever he used terms which were meant to facilitate understandings of his methodology, like 'scientific fantasies', 'intermediate representations', and 'scientific myths'.

How can we measure the relevance of Helmholtz's theories of energy to the Freudian concept of affect, or the contribution of Darwin's theory of evolution to the whole of Freudian thought, or the somewhat paradoxical usefulness of Lamarck's (1915-1985) ideas to Freud's *Übersicht der Übertragungsneurosen*²? The list could be much longer.

²A phylogenetic fantasy: overview of the transference neuroses

If, as mentioned, theories are constructed from the material available from outside knowledge, the theorists' choices, and are conditioned by the general knowledge available at the time, theories are also personal and subject to a variety of conscious and unconscious motivations. The importance and relevance of one theory or discipline to another cannot be determined and resolved by following a linear pattern.

In the aforementioned Symposium I also argued that:

Whoever reads the first chapter "Relations entre la science du langage et les autres sciences" in Roman Jakobson's (1963) book *Essais de linguistique générale* will have a good idea of what I am suggesting. It is a chapter with 300 bibliographic items, in which Jakobson examines a variety of disciplines with curiosity and enthusiasm: from the modern version of psychology of language based on the old German Sprachpsychologie, to genetics, Jakobson suggests a structural parallel between language and genetic code. He also turns to physics and to Niels Bohr's visit to the Massachusetts Institute of Technology where Jakobson and Bohr debated the relationship and differences between physics and linguistics, concluding that the opposition between them is unjust [...]

The originality of a theory can rest upon the combination of concepts which are already known and sometimes even not apparently relevant to the area under consideration. The outcome of the theory will depend to a great extent on the creativity of the theorist [...]

In conclusion, I believe that the relevance of one discipline to another cannot be entrusted to a linear evaluation - whether we are referring to a specific example (say one particular discovery) or more generally, or whether we are looking at the identity of the material taken into consideration (e.g. memory at molecular level or at psychoanalytic level), or the languages, methodologies, or epistemologies of the respective disciplines. The heuristic value that one discipline can have for another is more comprehensive, global and unpredictable. (Canestri, 2014)

It is necessary, moreover, to recognise in a discipline the different levels at which derivatives of other disciplines may or may not be used. I have mentioned elsewhere (Canestri, 2005) the example of Noah Chomsky's (2000) theory of language (I-language) as a cerebral organ, intended by the linguist as having a biological root. Chomsky (2000) himself, however, repeatedly highlighted the incongruity of using the molecular level if our aim is to carry out a discourse analysis. The same could be argued for the anthropologist Ian Tattersall's (2009) hypothesis of the existence of a gene carrying symbolic capabilities, or for the neurolinguist Andrea Moro's (2008) ideas about the neurobiological substrata underpinning 'impossible' grammars, carried out through neuro-imaging technics (magnetic functional resonance). Does this mean that all of this data is irrelevant to the linguist? Not at all, it only means that we have taken the wrong level of intersection, but this tells us nothing about the heuristic value of the data itself.

I would argue that what is valid for linguistics is valid for any other discipline, and certainly for the intersection between psychoanalysis and neuroscience. I am convinced of the relevance of neuroscience for psychoanalysis if, and only if (a strong conditional), proper consideration is given to the

issues highlighted so far, and to the one that I will go on to articulate. The complexity of the debate and the issues under scrutiny, however, should not discourage us from engaging in this dialogue, nor lead us to deem it irrelevant.

A well-known argument is the one concerned with the easy isomorphism of language; this can become a particularly pernicious issue in the interface between disciplines. There is always a question of translation in the different uses of the same words. Regarding the different employments of the (apparently) same language, the ‘radical interpretation’ of the philosopher of language Donald Davidson (2003) is a useful perspective to consider. Davidson (2003) postulates that the problem of interpretation is present even between speakers of the same language. A meaningful example could be the different employments of the word “representation” (*Vorstellung*) across different disciplines, even when the same word is used.

A final consideration, keeping to the area of enquiry explored above, is the fact that any interpretation of psychic phenomena inherently stands upon a general theory of the psyche. This is undoubtedly true for psychoanalysis. Upon the consideration and study of post-Freudian psychoanalytical models, it is possible to compare the extent of each contribution and conceive, as Arnold Cooper (1989) does, that one model might be better understood as more like a piece of a puzzle that is wrongly assumed to be a totality.

What is true for psychoanalysis can also stand for neuroscience. Many neuroscientific researches are extremely specific and understandably dig into the minute detail. The interest that neuroscience holds for psychoanalysis could be linked to the comprehensive vision that it offers. This perspective is in my perception shared by Yovell, Solms and Fotopoulou.

The ontological issue

In relation to the ontological issue, it is important to define the starting premises or assumptions. First of all, clarifying the elements at play implies considering two alternatives, broadly well-known and much debated: the dualist position and the monadic position.

The Philosopher Donald Davidson (1980) suggests the following specifications:

- a) nomological dualism, “which affirms that there are correlating laws” (Davidson, 1980:213-214) between the physical and mental events [interactionism, epiphenomenalism]
- b) nomological monism, “which affirms that there are correlating laws and that the events correlated are one” (Davidson, 1980:213-214) [materialism]
- c) anomalous dualism, “which combines the ontological dualism with general failure of laws correlating the mental and the physical” (Davidson, 1980:213-214) [Cartesianism]
- d) anomalous monism, which “claim that all events are physical, but rejects the thesis [...] that mental phenomena can be given purely physical explanations” (Davidson, 1980:213-214).

Davidson proposes three principles related to these specifications:

- 1) Principle of Causal Interaction: “it asserts that at least some mental events interact causally with physical events” (Davidson, 1980:208) (and vice versa)
- 2) Principle of the Nomological Character of Causality: “where there is causality, there must be a law” (Davidson, 1980:208)
- 3) Anomalism of the Mental. “There are no strict laws on the basis of which mental events can be predicted and explained” (Davidson, 1980:209)

Davidson argues that the inconsistency between these three principles – considering all of them to be true – is apparent. Moreover, he says that he accepts the identity between mental and physical events, but does not accept the idea “that a pure physical predicate, no matter how complex, has, as a matter of law, the same extension as a mental predicate” (Davidson, 1980:215).

The subsequent step states that “the mental is nomologically irreducible: there may be true general statements relating the mental and the physical, statements that have the logical form of a law; but they are not law-like” (Davidson, 1980:216).

In everyday practice, even in scientific practice, we use summary generalizations, or rather approximate generalizations, inasmuch as we do not know precise laws by explaining them.

But these generalizations are of two different classes:

- a) The generalization can be improved by resorting to the same vocabulary used in the first generalization, a vocabulary that corresponds to the final law and to one’s own conceptual domain. This generalization is called homonomic. A fitting example is given by physics.
- b) The generalization obtained leads us to believe that there is an operating law, but in order to enunciate it we have to resort to a different vocabulary. This is the heteronomic generalization.

From what has been said above, it is clear Davidson thinks that the general statements linking the mental and the physical are heteronomic, or rough heteronomic generalizations.

If this is true, it is not possible to accept the existence of strict psychophysical laws, inasmuch as languages and conceptual domains are dissimilar (categorical difference). If we accept these premises, the third principle mentioned above – the principle of the Anomalism of the Mental – is the logical conclusion.

This conclusion emphasises the identity of the mental and the physical, but also the fact that there are no strict laws at all which enable us to predict or to explain the mental phenomena.

Even knowing all the physical processes, it is necessary to have a pure psychological schema to interpret behaviours, affects, wishes, fantasies, etc.

To a mental event *m1* corresponds a physical event *p1* (bijective correspondence), but there is no isomorphism (Canestri, 2014).

In the chapter “*Mind and Brain. How do they relate?*” from *The Brain and the Inner World*, Mark Solms and Oliver Turnbull (2002) confront the mind-body problem, postulating a Kantian philosophical position regarding the limits of knowledge: they think that the nature of the relationship between brain and mind is not amenable to scientific proof. They claim the existence of a “Dual aspect monism” which, from my own point of view, is not so far from Davidson’s claim. I quote from their book:

... we are made of only one type of stuff (monist position) but this stuff is perceived in two different ways (hence dual-aspect monism); “... in our essence we are neither mental nor physical beings - at least not in the sense that we normally employ these terms [...]. This distinction between body and mind is therefore an artefact of perception, we can never escape the artificial mind-body dichotomy.

(Solms&Turnbull, 2002:56)

Therefore, what answer can we propose to the ‘ontological’ question of whether the entities described by the various disciplines coincide?

From a strictly ontological point of view the answer could be yes, as far as we accept the first Principle of Causal Interaction (Davidson, 1980), which affirms the identity between mental and physical phenomena. This is the Ontological Monism, materialism, which Freud also argued for.

But from an epistemological point of view, the consideration of the second Principle of the Nomological Character of Causality (Davidson, 1980), insofar as we cannot enunciate strict laws but just rough heteronomic generalizations, compels us to admit a descriptive dualism.

Therefore, the general conclusion says that the principle of the Anomalism of the Mental (Davidson, 1980) makes a claims for an ontological monism and an epistemological dualism (descriptive, methodological, and conceptual dualism).

With regards to the starting assumptions and the ontological issues, Item 1.3 in the paper I am commenting on here – “Recent Scientific Developments and their Impact on the Reception of Neuropsychanalysis in the Neurosciences” – proves particularly interesting.

It in fact allows us to glimpse a shift in the starting assumptions pertaining to previous research in neuroscientific, psychological, cognitive, neurophilosophical, neurobiological and genetic fields; a shift marked by a welcome abandonment of functionalism, computationalism and reductionism, opening the way for integrative and functional dynamic approaches. See also in this regard Item 3.2.1.1.: “Functionalism and the hardware-software fallacy”.

The functionalist approach is in fact contrary to everything we know around brain functioning and the body-mind relationship, not only from the vantage point of contemporary evidence, but even from Descartes’s work, as I propose to argue in the following.

The body and the meaning

The authors explore the body-mind relationship not only from the ontological point of view, but also from the perspective of the meaning that the body takes in the life of the mind and in the representation, or non-representation, of the drives and the damaged body. I would like at this point to consider an author who has been considered the representative par excellence of what Davidson called “anomalous dualism”, namely Descartes.

If we refer back to his book *Méditations métaphysiques* (1647) and we pause at his “Sixth Meditation”, subtitled “Of the Existence of Material Things, and of the real distinction between the Soul and Body of Man”, we may I believe find some surprises and clarifications on this topic.

Let’s consider the following short sentence where Descartes highlights:

And first of all there is no doubt that in all things which nature teaches me there is some truth contained [...] Nature also teaches me by these sensations of pain, hunger, thirst, etc., that I am not only lodged in my body as a pilot in a vessel, but that I am very closely united to it, and so to speak, so intermingled with it that I seem to compose with it one whole.³

(Descartes, 1911:29)

This certainly cannot be equal to a denial in Descartes of the dualist conception of mind and body, nor a repudiation of the distinction which attributes to the spirit the capacity to bear truth to matter. However, we find in this passage an underlining of the tight ‘conjunctum’ (connection) in which body and spirit find themselves. Not splitting, but ‘mêlange’, ‘permixtum’. And we also find highlighted how what stems from the body finds representation and meaning.

To the contrary, a computationalist premise would lead to the conclusion that a cerebral damage (and not only that) could not be processed and acquire meaning. This is proven wrong by experience, and evidence in this sense is strong. Equally strong is the evidence in the opposite direction, namely that meaning, the ‘talking cure’, modifies the brain. To deny this evidence means denying the therapeutic capabilities of psychoanalysis. Why was Freud neurologically right in proposing a ‘talking cure’? The authors of the paper summarise the answer by stating that “the brain shapes meanings and the meanings shape brain”, a concept clearly implied in Emily Dickinson’s poem.

The psychoanalytic model of the mind, as argued by the authors, “is embodied” (and also embedded). Neuroscience has afforded psychoanalysis contributions that have helped clarify and explain certain mental processes, and innumerable examples of these contributions can be found both in the in-depth exploration of theory and in clinical practice.

³“Docet etiam natura per istos sensus doloris, famis, sitis, etc., me non tantum adesse meo corpori ut nauta adest navigio, sed illi artissime esse conjunctum et quasi permixtum, adeo ut unum quid cum illo componam” (available at <http://www.thelatinlibrary.com/descartes/des.med6.shtml>)

First and foremost, I would like to quote the contributions afforded to the field of understanding and eventually treating post-traumatic syndromes: how trauma changes the brain, hippocampal atrophy, research on memory, on different types of memory, the lack of explicit memory with retain of implicit recall; child amnesia, implicit childhood memory; mnemonic defects; dissociative defences (Schacter, 2002); repression (competing information registered in long term memory can have an inhibitory effect on a specific memory) (Schacter, 2002); action of the right frontal lobe to avoid verbalization; how to investigate non conceptual and non-linguistic memories. In psychoanalysis we use countertransference, reverie, holding, empathy, non-verbal communication; false memories, etc.

The list cannot however leave aside, even in its brevity, the central contribution afforded by neuroscience to the developmental effects linked to trauma and stress, even in the foetus, which can be long-term: in the hypothalamus, fear-signalling impulses activate both the sympathetic neuron systems and the modulating systems of the HPA axis (Schechter et al., 2015). The HPA axis is involved in the neurobiology of mood disorders and functional illnesses, including anxiety disorder, bipolar disorder, insomnia, post-traumatic stress, borderline personality disorder, ADHD, major depressive disorder, alcoholism, etc. Several monoamine neurotransmitters are important in regulating the HPA axis, specially dopamine, serotonin, and norepinephrine. Increase in oxytocin + social interaction counteracts stress (Schechter et al., 2015).

The same could be said with regards to experiences and research on child-rearing (Gunnar, 2006) and cross-fostering (Weaver et al., 2004). Experiences in animals related to maternal behaviour and the relationship between genes and environment. Maternal behaviour actively stimulates hippocampus synaptogenesis ((Weaver et al., 2004).

In this, like other subject-matter, it is fundamental that we are aware of the key function that environmental factors play in soothing the potentially traumatic effects of early stress, and how this applies to the therapeutic role played in psychoanalysis.

In a recent book, *Finding the Body in the Mind. Embodied Memories, Trauma and Depression*, Marianne Leuzinger-Bohleber (2015) investigates many of these issues. Evidence from research on depression and trauma, and the frequently traumatic origins of chronic depression, illustrates very convincingly the importance that neuroscientific insights can have in contributing to think about working clinically with these pathologies.

In conclusion, the dialogue between neuroscience and psychoanalysis, avoiding any temptation towards reductionism and easy and false simplifications, is appealing from two perspectives: because it helps us reflect upon the consistency, pertinence and plausibility of our theories, deepening in many instances our understanding of mental phenomena. Secondly, because it affords in some cases (trauma, chronic depressions, etc.) some tools to better orient our therapeutic intervention. I do not believe this is negligible, nor do I believe that only neuroscience may be helpful in this sense. Many other disciplines (linguistics, semiotics, physics, maths, philosophy, etc.) may also contribute to psychoanalysis.

The opening of a dialogue in the intersections between disciplines favours heuristics.

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